Alief Randiansyah Pradanaputra

Data Analysis | Applied AI | Python Engineering

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Portfolio | GitHub | Kaggle | LinkedIn

Profile

Mathematics graduate who builds data pipelines, analytical tools, and AI applications from source data to working interfaces. Helped launch SEVIMA's K-12 program through its initial data work and built a rule-based validation system for hundreds of thousands of airline records. Combines business analysis with hands-on Python development, model serving, and automated testing.

Experience

Pusdatin Kementerian Perhubungan RI Sep 2026 - Present

Pranata Komputer Intern | Jakarta

- Built the ETL pipeline and dashboard processing hundreds of thousands of passenger records from more than 60 airlines; implemented 23 validation checks based on the ministry's reporting guideline.
- Served per-airline and per-day quality scorecards through FastAPI. Designed a production architecture with a scheduled worker, MySQL storage, and monthly ingestion; documented system design and an operational runbook.

SEVIMA Feb 2025 - Aug 2025

Data Analyst Intern | Surabaya

- Helped launch the K-12 expansion program as a core contributor to its initial data work. Built tuition-fee analysis and machine-learning customer scoring supporting acquisition of the program's first five school customers.
- Scraped and consolidated school and PDDIKTI data for analysis and modeling; analyzed active-user thresholds on the SiAkad platform.

Selected Projects

Tanya Sekolah — *Python, FastAPI, SQLite, ChromaDB, CatBoost*

- Built a natural-language application over 2,269 schools, combining guarded read-only SQL, multilingual RAG, and tuition estimation. Added Docker packaging and tests for data preparation, routing fallbacks, and API behavior.

Self-Retraining Model Platform — *MLflow, Argo Workflows, Kubernetes, PostgreSQL*

- Connected ingestion, training, evaluation, and serving in a local containerized platform. Implemented drift-triggered retraining, evaluation-gated promotion, model swaps without API redeployment, and automated checks.

Lights vs Numbers — *pandas, Plotly, Streamlit*

- Cleaned and aligned shipping activity, search trends, and official GDP series across time frequencies; built an annotated dashboard with traceable sources and explicit sample/live-data labels.

Education & Recognition

Institut Teknologi Sepuluh Nopember (ITS) Aug 2022 – Aug 2026

Bachelor of Mathematics (S.Mat)

Top 20, Microsoft AI Impact Challenge Datathon 2026 — DAWG-ID, a macro-shock vulnerability portal covering 38 Indonesian provinces.

ITS Team Lead, Satria Data Big Data Challenge (Aug–Sep 2025); previously taught first-year programming as a laboratory assistant at ITS (Aug–Dec 2024).

Skills

Data & analysis: Python, pandas, SQL, ETL, PostgreSQL, SQLite, Google Sheets, Selenium, Plotly, Streamlit

AI & engineering: LLM APIs, RAG, ChromaDB, CatBoost, PyTorch, FastAPI, Docker, Kubernetes, MLflow, Argo Workflows, DVC, pytest, GitHub Actions

English: EFL score 567